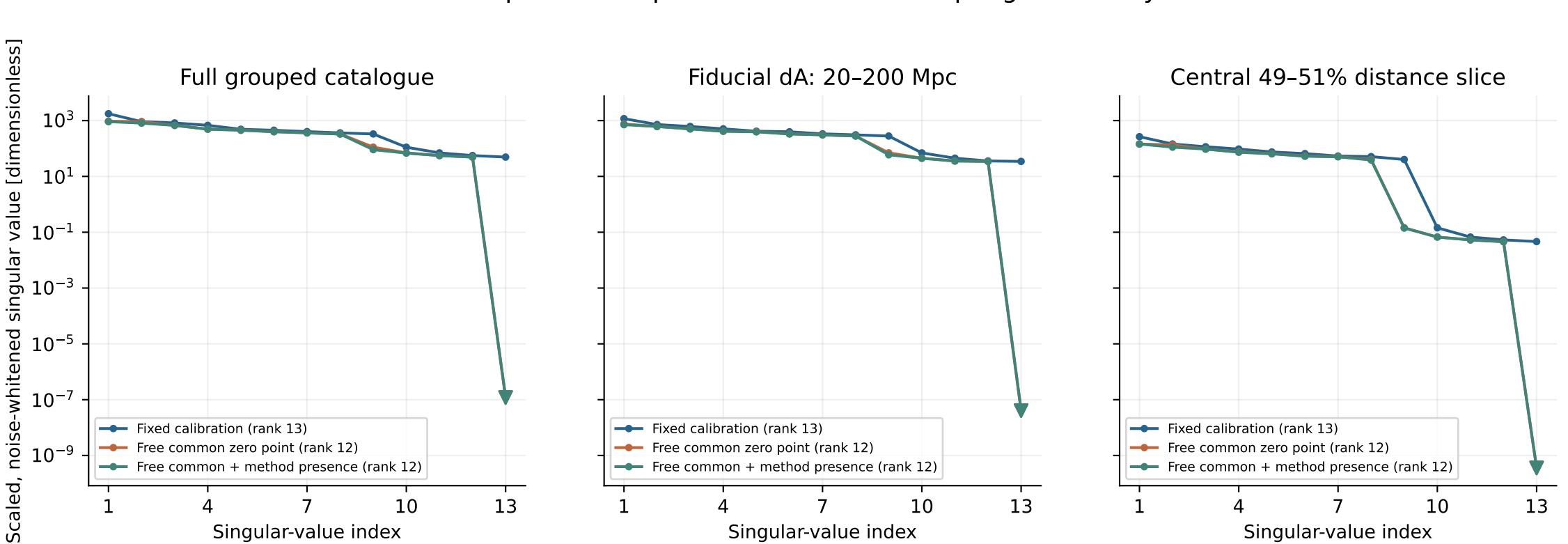

Published modulus errors; scenario velocity/motion errors = 100/300 km/s. Same fixed coefficient scales in every panel.

Downward triangles: singular values at/below numerical rank tolerance, displayed at that tolerance (raw values retained).

Reused R2-02 SVD; no response fit. Free common calibration leaves one unestimable direction; a full-rank scenario is not a detection.